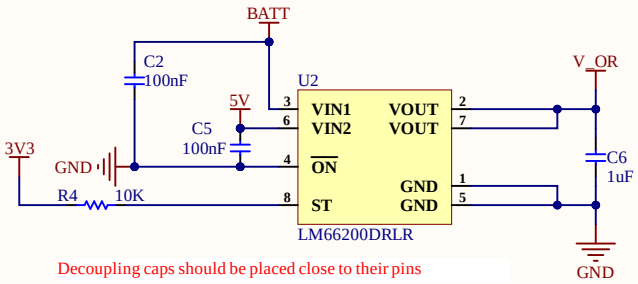


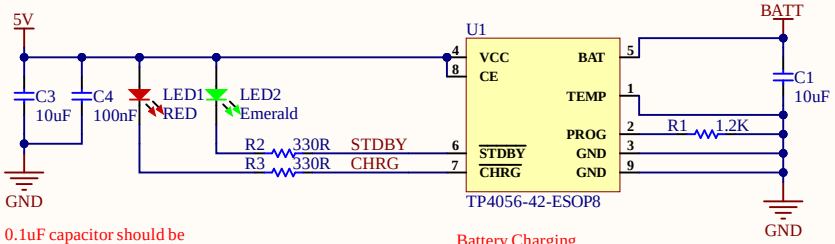
POWER SUPPLY ORING CIRCUIT



Decoupling caps should be placed close to their pins

The highest voltage supply is automatically selected. ST pin is used to detect which voltage is currently selected

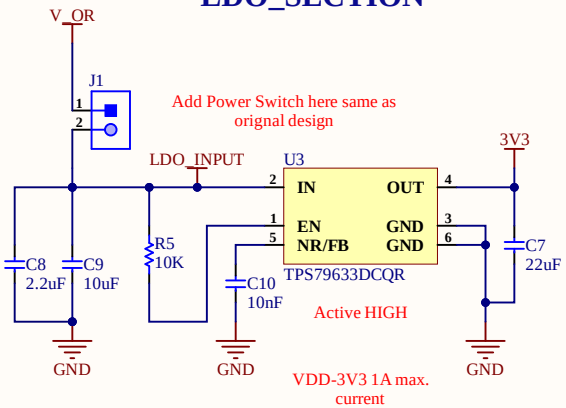
BATTERY CHARGING CIRCUIT

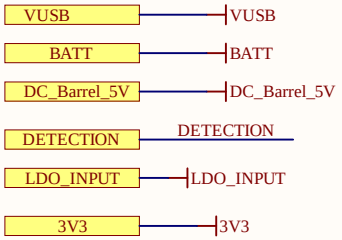


0.1uF capacitor should be placed closest to VCC pin

Battery Charging current = 1000mA

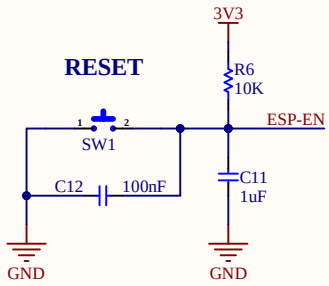
LDO_SECTION

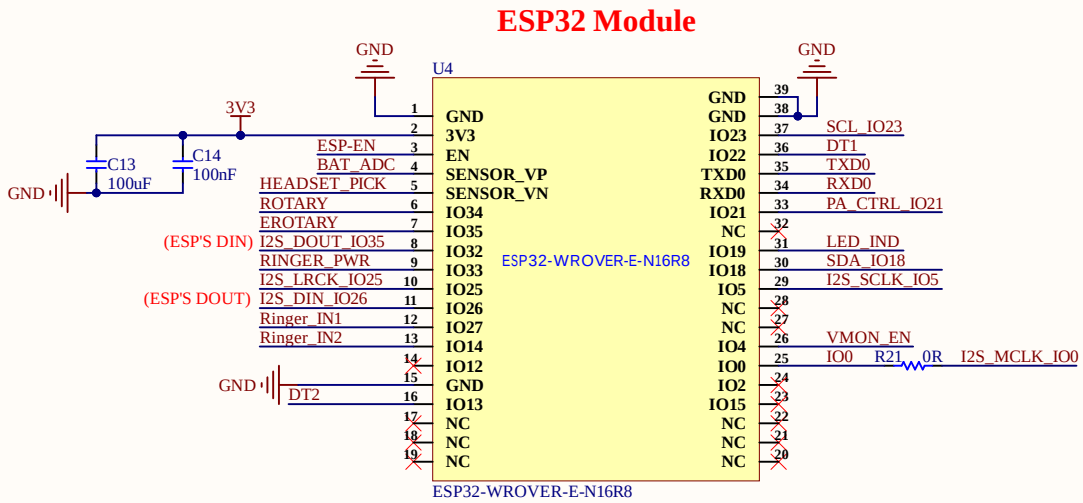


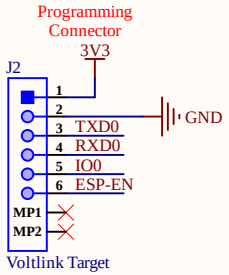


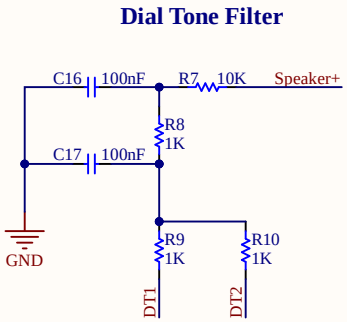
eSpark Consulting Group

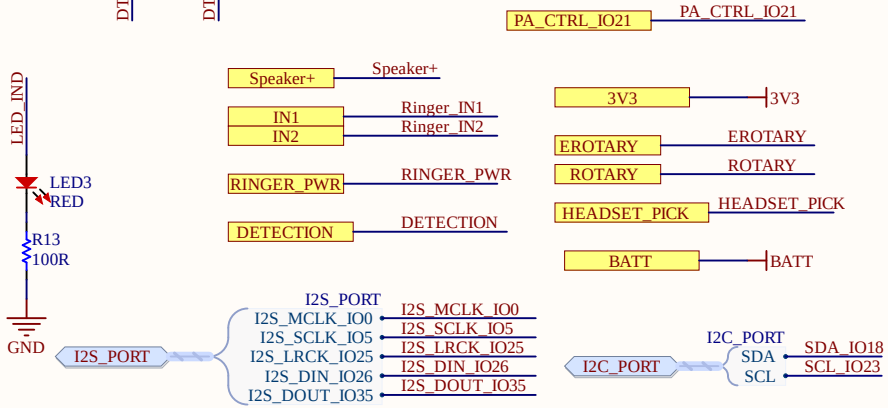
Title: Power_Sheet.SchDoc	
File: Telephone Circuit V1.0.PrjPcb	
Date: 27/01/2025	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 6

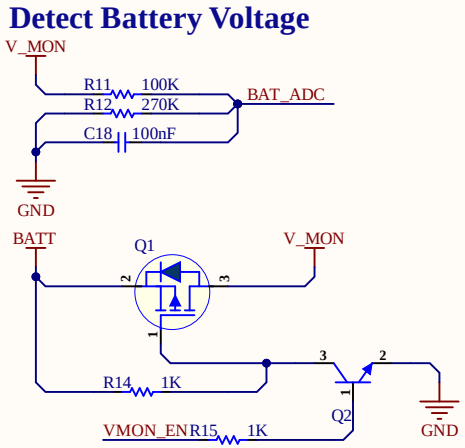






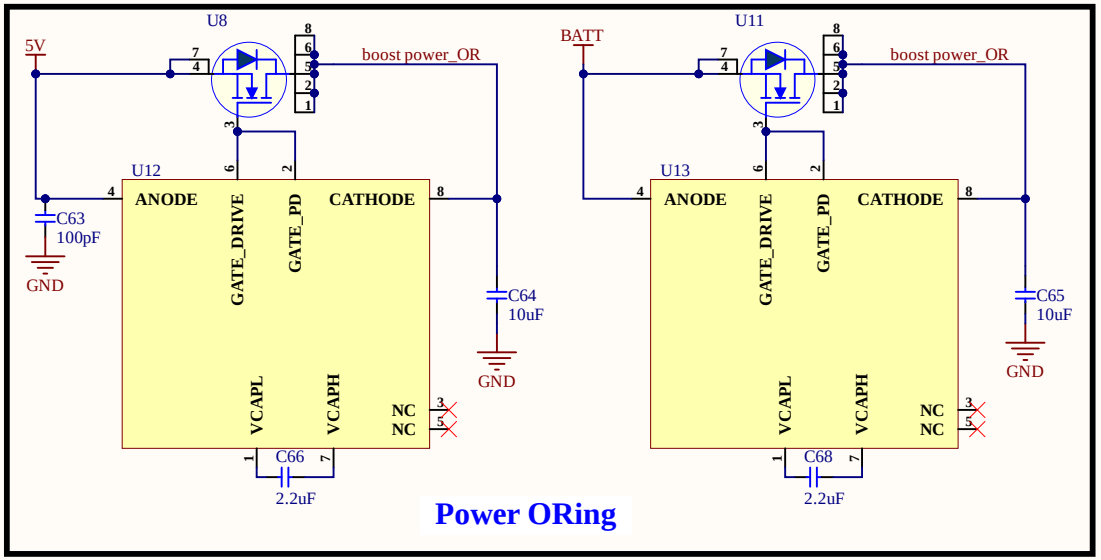


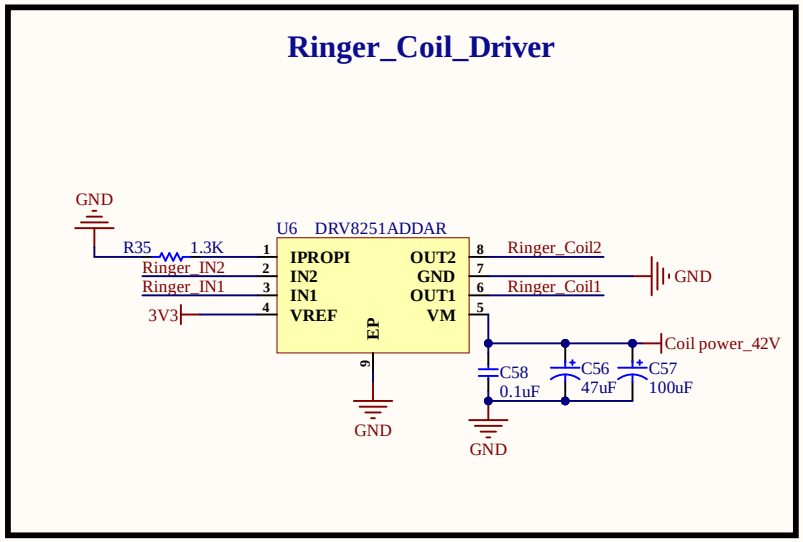


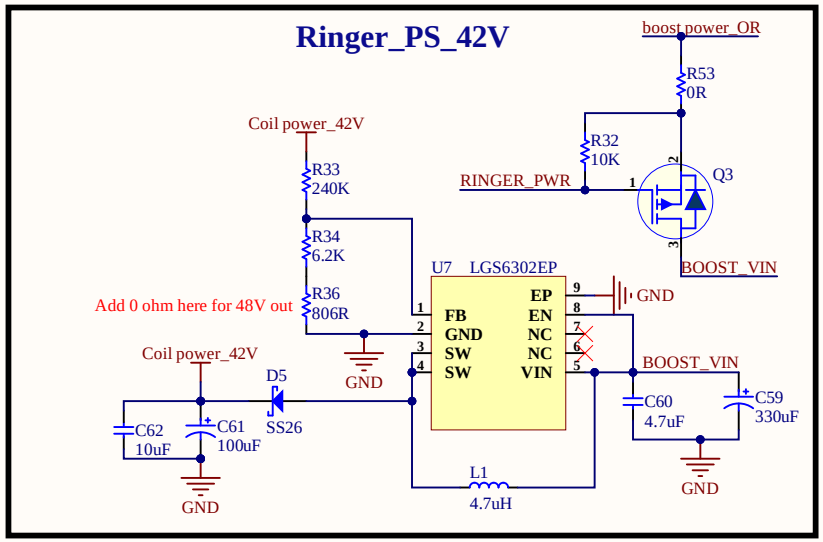


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Title: Controller_Sheet.SchDoc	
File: Telephone Circuit V1.0.PrjPcb	
Date: 30/01/2025	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 6







3V3 — 3V3

BATT — BATT

RINGER_PWR — RINGER_PWR

IN1 — Ringer_IN1

IN2 — Ringer_IN2

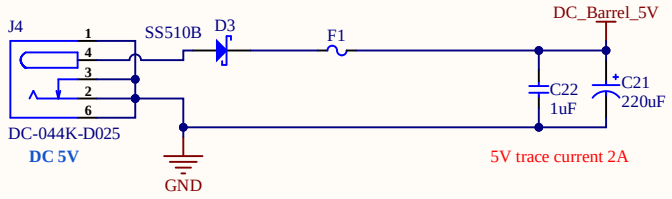
Ringer_Coil1 — Ringer_Coil1

Ringer_Coil2 — Ringer_Coil2

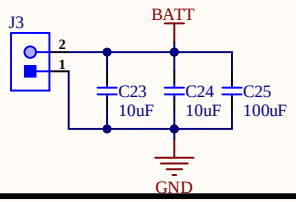
eSpark Consulting Group

Title: Ringer_Sheet.SchDoc	
File: Telephone Circuit V1.0.PrjPcb	
Date: 30/01/2025	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 6

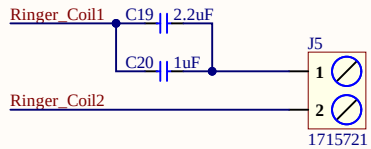
DC BARREL JACK INPUT 5V



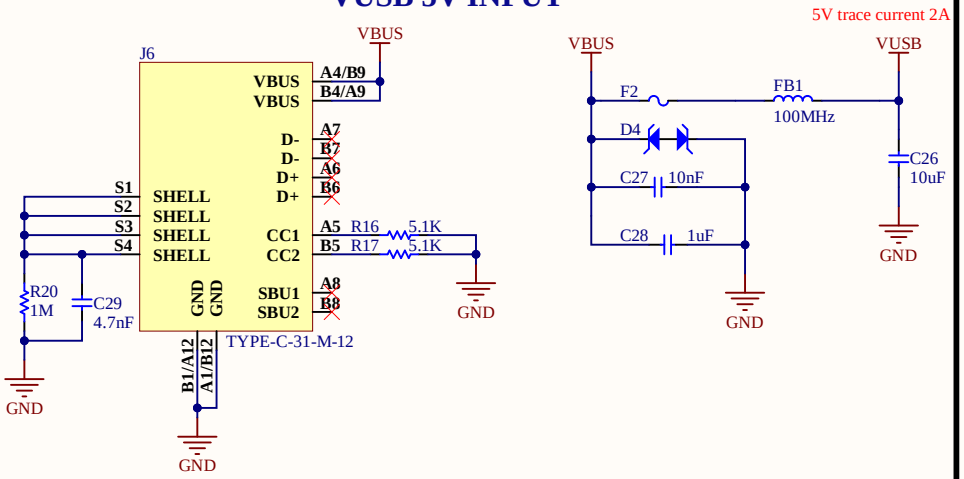
BATTERY INPUT



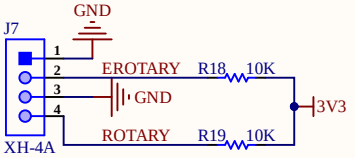
Ringer_Coil



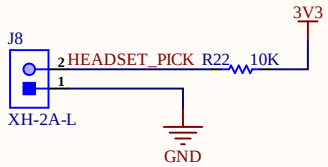
VUSB 5V INPUT



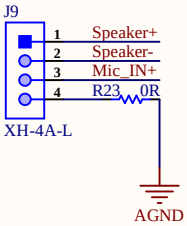
Rotary Dialer



Headset Detection



Headset_Mic and Speaker



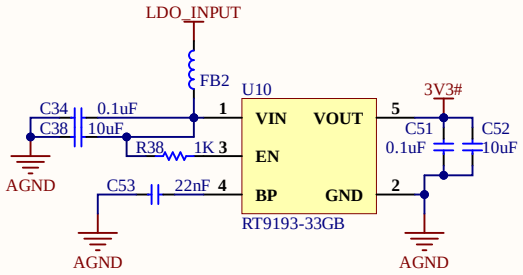
EROTARY
ROTARY
HEADSET_PICK
VUSB
BATT
DC_Barrel_5V
Ringer_Coil1
Ringer_Coil2
3V3

Speaker+
Speaker-
Mic_IN+

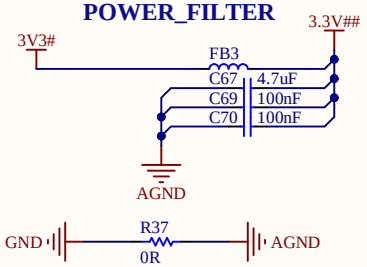
eSpark Consulting Group

Title: Connector_Sheet.SchDoc	
File: Telephone Circuit V1.0.PrjPcb	
Date: 30/01/2025	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 6

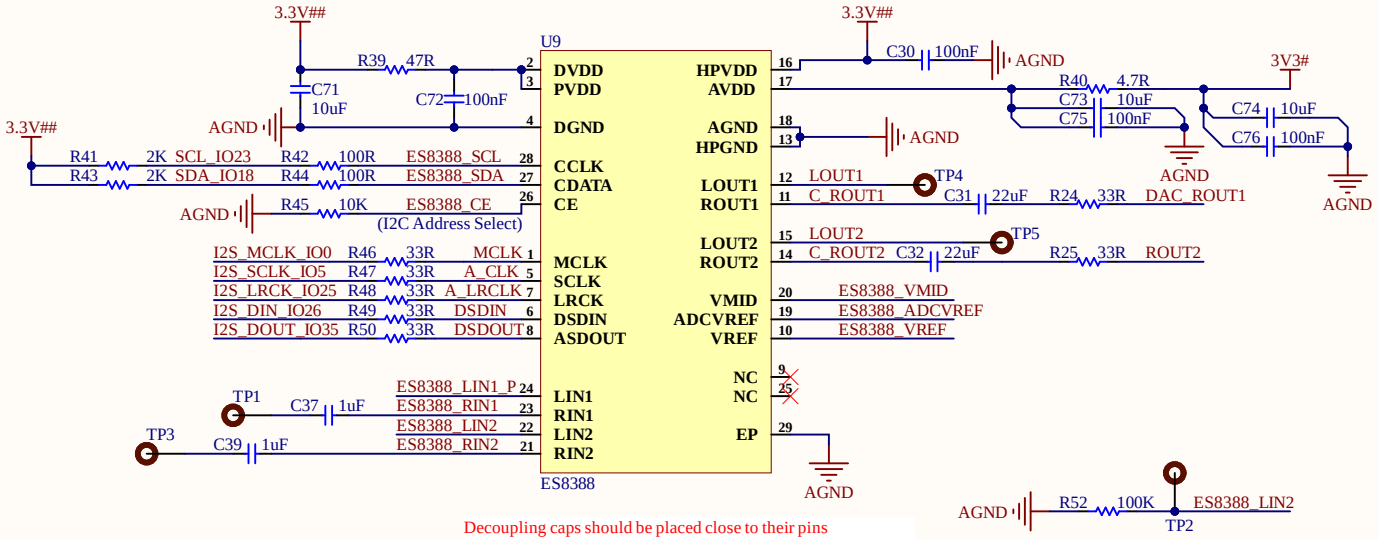
AUDIO_LDO_SECTION



POWER_FILTER

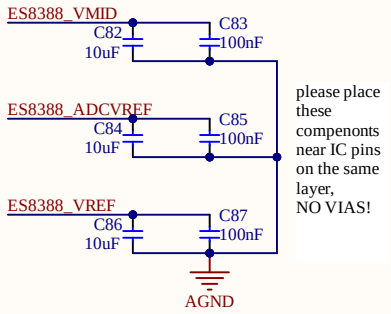


Audio Codec



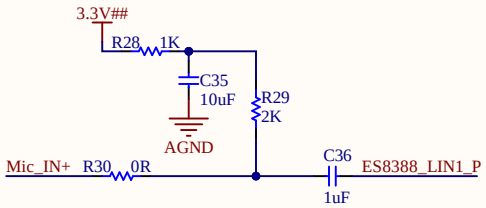
Decoupling caps should be placed close to their pins

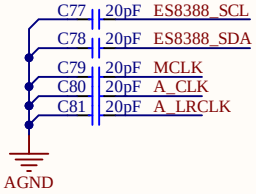
SUPPLY_DECOUPLING

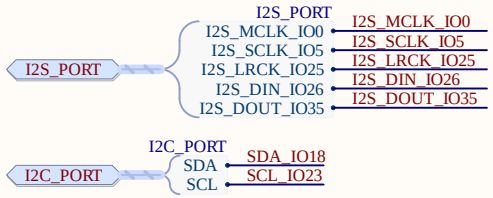


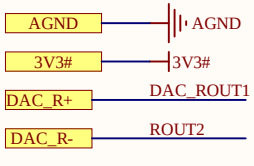
please place these compenonts near IC pins on the same layer, NO VIAS!

MIC Section







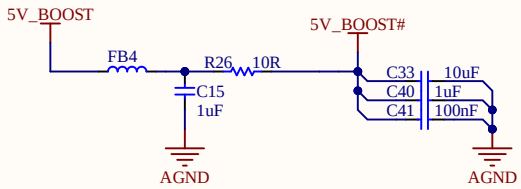


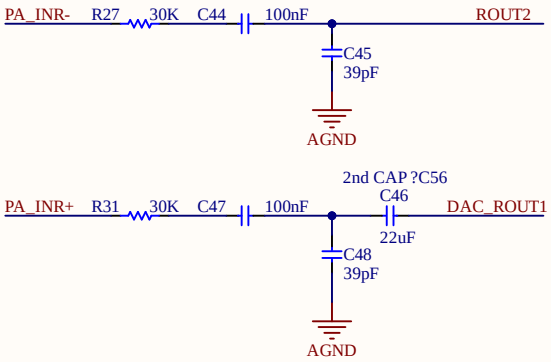


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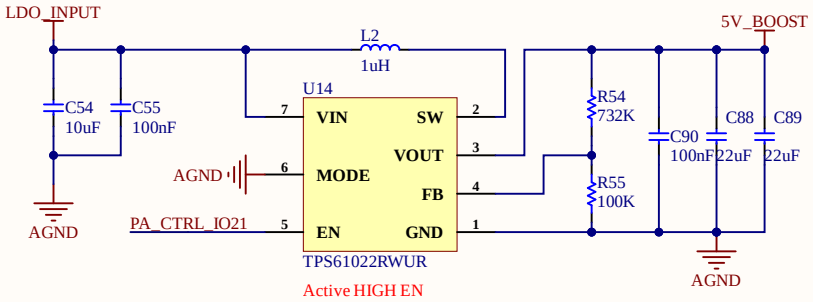
Title: Audio Codec_Sheet.SchDoc		
File: Telephone Circuit V1.0.PrjPcb		
Date: 27/01/2025	Page	Num: 5
Drawn By: Huzaifa Ali	Page	Total: 6

Power_Filter

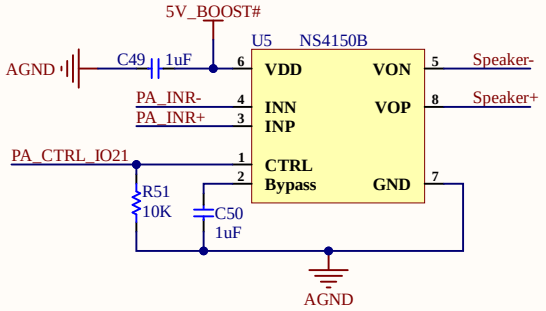


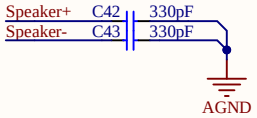


BOOST CONVERTER



Audio Amplifier





LDO_INPUT — LDO_INPUT

AGND — AGND

DAC_R+ — DAC_ROUT1

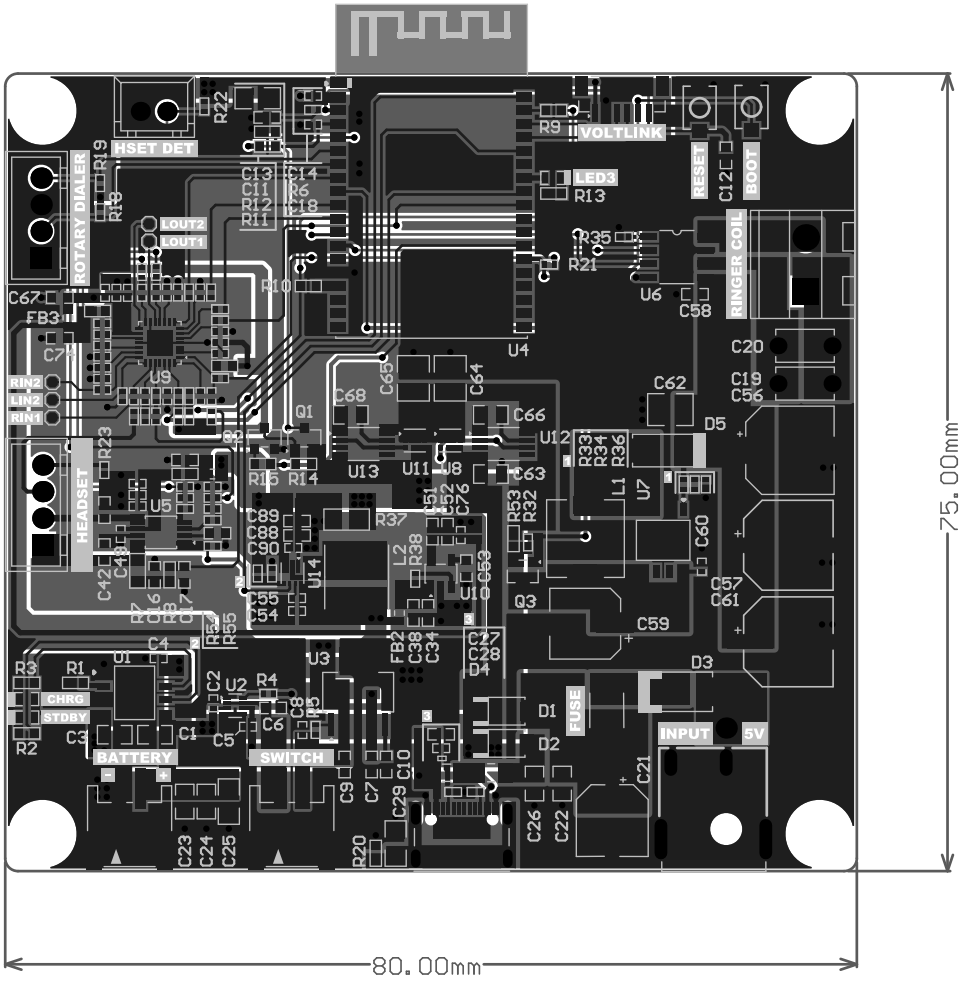
DAC_R- — ROUT2

Speaker+ — Speaker+

Speaker- — Speaker-

PA_CTRL_IO21 — PA_CTRL_IO21

Title: Audio Amplifier_Sheet.SchDoc	
File: Telephone Circuit V1.0.PrjPcb	
Date: 30/01/2025	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 6



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.010mm	3.5	
1	Top Layer		0.036mm		
	Dielectric 1	FR-4	1.500mm	4.8	
2	Bottom Layer		0.036mm		
	Bottom Solder	Solder Resist	0.010mm	3.5	
	Bottom Overlay				
Total board thickness:			1.591mm		